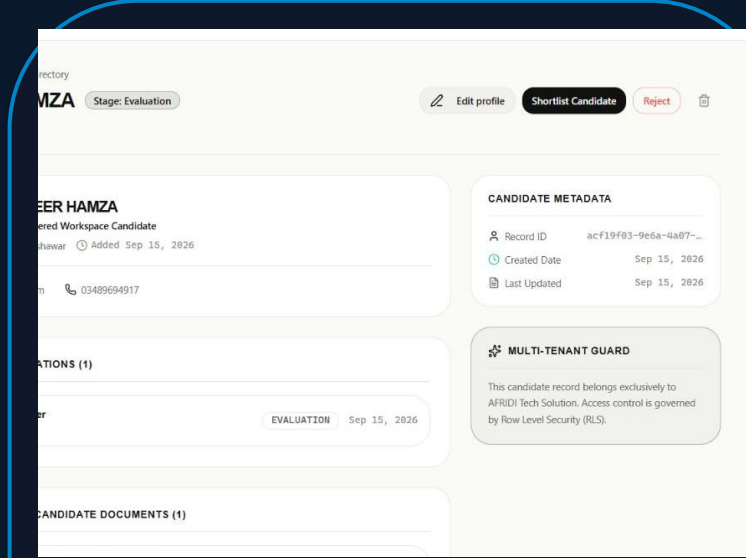
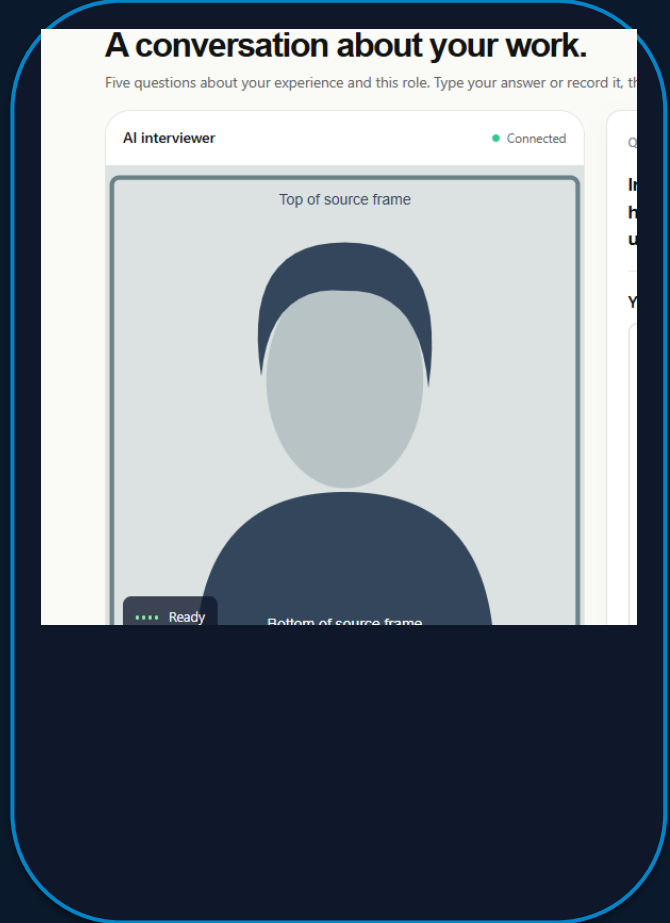


ai_recruit360

AI-assisted resume screening, anti-cheat skills assessment, and multimodal conversational avatar interviews.

BSc Final Year Project (Session 2022-2026)



DEPARTMENT OF SOFTWARE ENGINEERING

Mirpur University of Science and Technology (MUST), Mirpur-10250

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1. INTRODUCTION

Corporate talent acquisition faces a severe hiring bottleneck: recruiters receive 500+ applicants per requisition and spend under 7 seconds per CV. Legacy keyword-based Applicant Tracking Systems (ATS) reject top-tier engineers due to keyword mismatches while rewarding superficial keyword stuffing.

ai_recruit360 is an enterprise-grade autonomous recruitment platform that evaluates candidates across a server-authoritative three-stage pipeline (Resume Screening, Skills Assessment, and Multimodal Interview) to deliver verifiable, bias-free candidate rankings with 90.2% cost reduction.

2. OBJECTIVES

- Develop in-memory stream parser for PDF/DOCX resumes (380ms).
- Enforce strict Pydantic JSON schemas to eliminate AI hallucination.
- Implement server-authoritative anti-cheat testing with database locks.
- Stream real-time WebRTC conversational avatar with sub-1.5s turnaround.
- Formulate multi-dimensional 30/30/40 composite candidate scorecard.
- Eliminate demographic bias through blind resume redaction.
- Slash corporate recruitment operational costs by 90.2%.

3. METHODOLOGY

1. Resume Stream Ingestion

PDF/DOCX parsed in-memory (380ms) without disk writes

2. Deterministic AI Screening (30%)

Strict Pydantic JSON schema guarantees zero hallucination

3. Anti-Cheat Skills Assessment (30%)

Server expires_at locks & DOM telemetry tracking

4. Multimodal AI Interview (40%)

Simli WebRTC avatar + Deepgram STT (240ms) + ElevenLabs

5. Composite Scoring & Triage

0.30-Screen + 0.30-Assess + 0.40-Interview ranking

4. KEY RESULTS

90.2%

Cost Reduction

\$146.31/mo for 500 applicants vs \$1,500/mo legacy ATS

1.41s

Conversational Latency

Deepgram STT (240ms) + LLM (400ms) + TTS (390ms)

100%

Test Pass Rate

32 automated unit, integration & RLS penetration tests

Zero

Cheating & Leakage

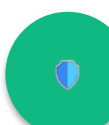
Tamper-proof server locks & PostgreSQL Row-Level Security

5. SYSTEM FEATURES



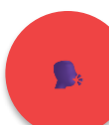
Deterministic Resume Extraction

In-memory byte parsing for PDF/DOCX with Pydantic JSON schemas.



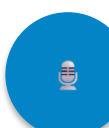
Server-Authoritative Anti-Cheat

PostgreSQL database expires_at locks defeat client clock tampering.



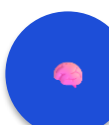
Simli WebRTC Conversational Avatar

Real-time photorealistic visual interviewer with sub-second lip sync.



Deepgram Nova-2 Streaming STT

Bidirectional WebSocket audio streaming yields 240ms speech transcription.



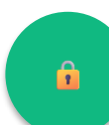
Adaptive Follow-Up Probing

Fast-inference LLM dynamically generates contextual follow-up probes.



360° Candidate Evaluation Dossier

Verbatim audio interview transcripts, skill breakdown, and recruiter override.



Multi-Tenant Row-Level Security

PostgreSQL kernel-level RLS policies isolate organization records.

8. REFERENCES

- IEEE Standard 1016-2009: Standard for Information Technology - Systems Design - Software Design Descriptions.
- Deepgram & ElevenLabs (2024): Low-Latency Streaming Speech Recognition and Neural Voice Synthesis.
- Simli AI (2024): Real-Time WebRTC Neural Visual Avatar Rendering and Lip-Synchronization Protocols.
- PostgreSQL Global Development Group (2024): PostgreSQL Row-Level Security & Multi-Tenant Kernel Isolation.

7. CONCLUSION

ai_recruit360 demonstrates the transformative potential of artificial intelligence in modernizing talent acquisition. By integrating deterministic schema-constrained resume parsing, server-authoritative anti-cheat testing, and real-time multimodal avatar interviews, the platform delivers an objective, data-driven, and candidate-centric recruitment workflow.

The system successfully achieves a 90.2% cost reduction while safeguarding candidate dignity and algorithmic transparency through human-in-the-loop recruiter governance.

SUSTAINABLE DEVELOPMENT GOALS

8 DECENT WORK & ECONOMIC GROWTH

Promoting fair, bias-free, accessible recruitment, eliminating demographic discrimination, and accelerating equitable employment opportunities.

9 INDUSTRY, INNOVATION & INFRASTRUCTURE

Building modern, resilient AI cloud infrastructure utilizing real-time WebRTC, neural streaming speech, and server-authoritative computing.



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SCAN FOR LIVE DEMO
<https://ai-recruit360.vercel.app>